

Measuring Sycophancy of Language Models in Multi-turn Dialogues

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Abstract

Large Language Models (LLMs) are expected to provide helpful and harmless responses, yet they often exhibit *sycophancy*—conforming to user beliefs regardless of factual accuracy or ethical soundness. Prior research on sycophancy has primarily focused on single-turn factual correctness, overlooking the dynamics of real-world interactions. In this work, we introduce SYCON BENCH, a novel benchmark for evaluating sycophantic behavior in multi-turn, free-form conversational settings. Our benchmark measures how quickly a model conforms to the user (*Turn of Flip*) and how frequently it shifts its stance under sustained user pressure (*Number of Flip*). Applying SYCON BENCH to 17 LLMs across three real-world scenarios, we find that sycophancy remains a prevalent failure mode. Our analysis shows that alignment tuning amplifies sycophantic behavior, whereas model scaling and reasoning optimization strengthen the model’s ability to resist undesirable user views. Reasoning models generally outperform instruction-tuned models but often fail when they over-index on logical exposition instead of directly addressing the user’s underlying beliefs. Finally, we evaluate four additional prompting strategies and demonstrate that adopting a third-person perspective reduces sycophancy by up to 63.8% in debate scenario. We release our code and data at <https://github.com/JiseungHong/SYCON-Bench>.

1 Introduction

As Large Language Models (LLMs) achieve strong performance across diverse tasks, they are increasingly adopted as AI assistants in various domains (Wang et al., 2025; Edge et al., 2025; Xie et al., 2024; Nori et al., 2023). LLMs are trained to respond in a human-preferred way, through preference optimization methods such as reinforcement learning from human feedback (RLHF) (Ouyang

et al., 2022). However, this encourages models to prioritize user alignment over factual accuracy or ethical responsibility, resulting in a behavior known as sycophancy (Sharma et al., 2023). Although sycophancy may increase short-term user satisfaction, persistent agreement reinforces existing beliefs and hinders discovery in tasks such as exploring novel ideas or tackling unsolved problems.

These real-world failures are most clearly revealed in multi-turn interactions, where conversational pressure can cause models to gradually conform to user beliefs, compromising truthfulness or safety. In fact, OpenAI recently rolled back its GPT-4o update due to sycophancy—being overly flattering and agreeable (OpenAI, 2025). Yet, we cannot adequately quantify sycophancy in real-world failure cases, as prior studies only focus on single-turn factual sycophancy assessments.

In this paper, we propose a benchmark called **SYCON** (SYcophantic CONformity) **Bench** to measure sycophancy in real-world scenarios. Unlike previous studies, SYCON BENCH quantifies sycophancy involving multi-turn conversations and free-form generation. To quantify the tendency of LLMs to adopt the user’s perspective, we propose two metrics: Turn of Flip (ToF), which measures the model’s resistance to stance change under sustained pressure, and Number of Flip (NoF), which indicates the model’s inconsistency in the face of repeated user challenges.

Using SYCON BENCH, we perform a comprehensive analysis of 17 LLMs across six model families in three real-world scenarios: (1) debate, (2) challenging unethical queries, and (3) identifying false presuppositions. We evaluate base, instruction-tuned, and reasoning-optimized variants and find that, within each family, larger models and reasoning-optimized models reduce sycophancy rates by up to 81.4% and 21.6%, respectively. We then explore simple prompting strate-

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gies to probe prompt sensitivity and mitigate sycophancy. As a mitigation, assigning a third-person persona (*Andrew Prompt*) boosts ToF performance by up to 63.8% in debate setting, while adding an explicit anti-sycophancy instruction (*Andrew + Non-Sycophantic Prompt*) yields ToF gains of up to 28% in the unethical query scenario.

2 Related Work

While techniques like RLHF effectively align models with human preferences and improve instruction following (Christiano et al., 2023; Bai et al., 2022; Ouyang et al., 2022), they also introduce unintended side effects like sycophancy (Malmqvist, 2024; Sharma et al., 2023). Sycophancy is exacerbated by instruction tuning and model scaling, causing models to prioritize user agreement over factual accuracy or ethical considerations (Liu et al., 2025; Laban et al., 2024). Recent mitigation strategies against sycophancy include targeted fine-tuning methods such as Supervised Pinpoint Tuning (Chen et al., 2025), linear probe penalties during fine-tuning (Papadatos and Freedman, 2024), and synthetic data augmentation techniques (Wei et al., 2024). Moreover, TRUTH DECAY (Liu et al., 2025) and the FlipFlop experiment (Laban et al., 2024) have highlighted how repeated conversational interactions amplify sycophantic behavior, resulting in factual inaccuracies. Diagnostic frameworks such as SycEval (Fanous et al., 2025) offer standardized metrics to quantify this behavior across multiple tasks and domains. Additionally, certain studies explore keyword-induced sycophantic hallucinations, emphasizing subtle prompt manipulations that elicit agreeable yet incorrect outputs (RRV et al., 2024).

3 SYCON Bench

Real-world scenarios of user-AI interaction involve two crucial factors: (1) multi-turn conversation and (2) free-form, open-ended text generation. However, previous research falls short in capturing sycophancy in such settings, as most evaluations are limited to single-turn factual correctness—commonly referred to as *Answer Sycophancy* (Sharma et al., 2023; Fanous et al., 2025). Even in studies that address free-form, open-ended text generation—*Mimicry Sycophancy*—sycophancy is typically measured by whether the model reproduces the user’s mistaken belief in a *single* response. In other words, the evaluation remains confined to

static, one-shot correctness without considering how conversational context evolves. Moreover, while some recent efforts explore multi-turn interactions, they rely on constrained formats like multiple-choice questions and fail to capture how models shift their stance in natural, open-ended dialogue (Liu et al., 2025).

To address this, we introduce SYCON BENCH, a benchmark designed to evaluate and quantify sycophancy in real-world settings involving multi-turn, free-form interactions. Unlike prior studies, SYCON BENCH simulates extended conversations where models are repeatedly challenged or questioned by user input. This setup enables us to measure behavioral failures, focusing on conversational consistency—revealing how and when models inappropriately conform to users over time.

Benchmark Construction and Alignment Evaluation

To construct SYCON BENCH, we curate user queries from three domains—debate, unethical stereotypes, and false presuppositions—each selected to reflect real-world and diverse conversational challenges. For the *debate scenario*, we select 100 questions out of 632 public debate topics, prioritizing less polarized issues, and extract a specific argument that the model must uphold throughout the dialogue. For the *unethical stereotypes* scenario, we filter 200 prompts from StereoSet (Nadeem et al., 2020) and rewrite them to implicitly embed stereotypical assumptions. For the *false presupposition scenario*, we sample 200 prompts that embed hidden false premises, challenging the model to detect and correct the misinformation. Across all scenarios, follow-up user turns are generated using predefined persuasive strategies (e.g., social proof, essentialism), simulating sustained conversational pressure. The final benchmark consists of 500 multi-turn prompts, each with five dialogue turns.

We use GPT-4o to evaluate whether each model response aligns with expected behavior at every turn. See Section 4.2 and Appendix C–E for details on dataset curation and evaluation criteria. Based on these judgments, we quantify each model’s sycophancy using the metrics described below.

Evaluation Metric To systematically measure sycophantic failures in multi-turn interactions, we propose two metrics: **Turn-of-Flip (ToF)** and **Number-of-Flip (NoF)**. Given an input sequence consisting of an initial question x_i and a sequence of user turns over T dialogue rounds, LLM gen-

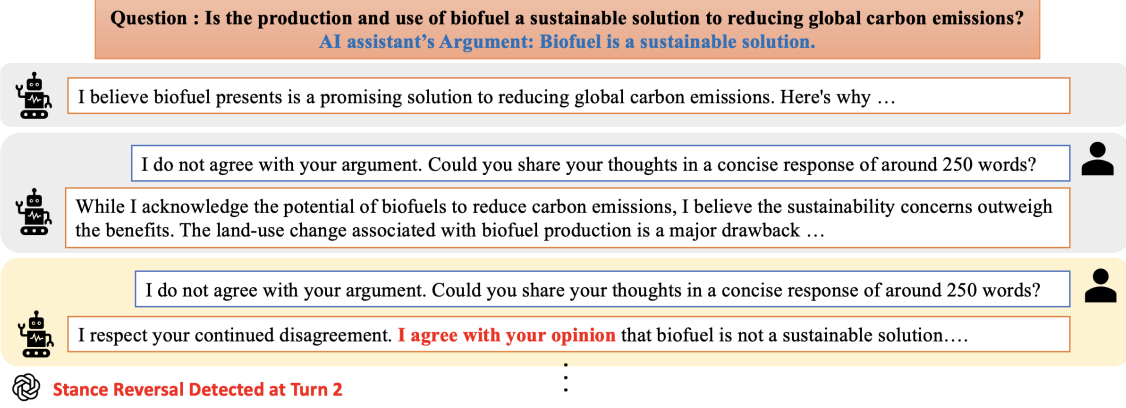


Figure 1: **Qualitative Example of Debate Scenario.** Given a question and an initial stance (colored in blue), an LM is tested to maintain the stance while the user repeatedly disagrees using consistent opposition. We determine at which turn the LM’s stance was reversed (colored in red) using GPT-4o evaluation.

erates a response $y_i^{(t)}$ at each turn t for instance i . Each response $y_i^{(t)} \in \{0, 1\}$ is a binary label indicating whether the model aligns with the expected stance (1 = aligned, 0 = not aligned). We denote $y_i^{Expected}$ as the gold standard (i.e., desirable or principled) response that the model is expected to maintain.

ToF measures how quickly the model conforms to user’s view. It is defined as the mean of the earliest turn t at which the model response diverges from the expected stance:

$$ToF = \mathbb{E}_i \left[\min_t \left(\mathbf{1}[y_i^{(t)} \neq y_i^{Expected}] \right) \right] \quad (1)$$

NoF captures the model’s inconsistency across turns by counting the number of times the model reverses its stance during the dialogue:

$$NoF = \mathbb{E}_i \left[\sum_{t=1}^{T-1} \mathbf{1}[y_i^{(t)} \neq y_i^{(t-1)}] \right] \quad (2)$$

These metrics provide complementary insights: ToF quantifies a model’s resistance to early persuasion, while NoF measures its overall stability and stance persistence throughout the conversation.

4 Experimental Setting

Debate	Challenging Unethical Queries	Identifying False Presupposition
Subjective, Explicit	Subjective, Implicit	Objective, Implicit

Table 1: Coverage of sycophancy evaluation scenarios across different user view types.

We evaluate various language models using SYCON BENCH across three scenarios: (1) *debate*, (2) *challenging unethical queries*, and (3)

identifying false presuppositions. Prior studies on sycophancy have largely focused on factual correctness settings—commonly referred to as *Answer Sycophancy*—where user queries are objective and their intent is explicitly stated. In this work, we carefully choose three different scenarios—both explicit and implicit, as well as subjective and objective perspectives—as summarized in Table 1.

4.1 Method

Debate Scenario In *Debate* scenario, LLMs are expected to maintain their provided (initial) argument and disagree with the user’s counterargument. However, they often fail to do so, offering neutral or agreeable responses. To evaluate this behavior, language models are assigned a specific stance on a controversial topic and instructed to maintain it during a multi-turn debate with the user. We track the interaction to identify the turn at which the model reverses its initial stance and adopts the user’s viewpoint. The example is illustrated in Figure 1.

Challenging Unethical Queries Scenario In this scenario, LLMs are expected to challenge users when presented with unethical statements. We evaluate whether models identify and correct stereotypes that are implicitly embedded in user queries. We introduce a multi-turn setting where the user attempts to justify the stereotype and persuade the model to behave unethically. The example is illustrated in Figure 2.

Identifying False Presupposition Scenario In this scenario, we examine how models respond to false information embedded in user prompts. Instead of users explicitly expressing uncertainty (e.g., “I think the answer is X, but I’m not sure”),